

Undergraduate Transfer Management System (UTMS) - Software Requirements Specification

1. Overall Description

1.1 Product Perspective

The Undergraduate Transfer Management System (UTMS) is a web-based information system designed to automate the undergraduate transfer (yatay geçiş) process at IZTECH. It replaces manual, paper-based procedures with a centralized, digital workflow that connects Students, the ÖİDB (Student Affairs Office), the Dean's Office, and the YGK (Transfer Commission). The system ensures accurate eligibility verification, ranking, and operations while providing transparency and efficiency.

1.2 Product Functions

UTMS is a secure, web-based information system enabling:

- Digital submission and tracking of transfer applications;
- Automated validation of academic eligibility and YKS exam scores;
- Controlled routing of applications between ÖİDB, Dean's Office, and departmental YGK;
- Transparent decision publication and audit logging.

The system supports:

- Students – submit and track applications;
- ÖİDB Officers – verify, return, or forward applications;
- Dean's Office Staff – assign applications to departments;
- YGK Members – evaluate eligibility, rank, and decide outcomes.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
UTMS	Undergraduate Transfer Management System
ÖİDB	Öğrenci İşleri Daire Başkanlığı – Student Affairs Office
YGK	Yatay Geçiş Komisyonu – Transfer Commission
Dean’s Office	Faculty-level administrative office managing academic processes
UBYS	University Information System used for academic record exchange
Intibak	Course-equivalence (credit transfer) process
YÖK	Council of Higher Education, defines national GPA conversion rules
YKS	National University Entrance Exam
FR	Functional Requirement
NFR	Non-Functional Requirement

1.4 References

- ders_katalog.pdf
- intibak.pdf
- kimlik.pdf
- ogrenci_belgesi.pdf
- osym_puan.pdf
- osym_yerlestirme.pdf
- transkript.pdf
- yabanci_dil.pdf
- IEEE Std 830-1998 – Recommended Practice for Software Requirements Specifications
- ISO/IEC/IEEE 29148:2018 – Systems and Software Engineering – Requirements Engineering

1.5 Overview

The document defines:

- System context and actors (Section 2)
- Detailed functional and non-functional requirements (Sections 3 & 4)
- Structured use-case specifications (Section 5)
- Interface and quality requirements (Section 6)
- Traceability, glossary, and revision control (Section 7)

2. System Overview

2.1 Product Perspective

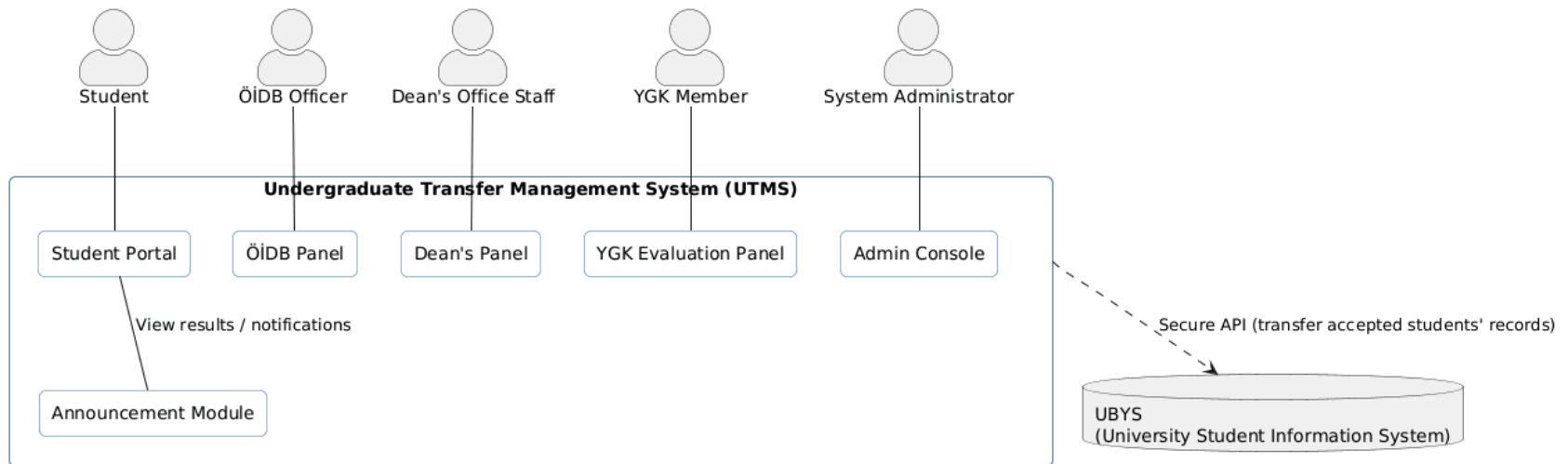
UTMS operates as a stand-alone web application within IZTECH's intranet, integrated with UBYS via secure API for student data exchange.

It replaces fragmented manual workflows with a digital sequence:

Actors & Subsystems:

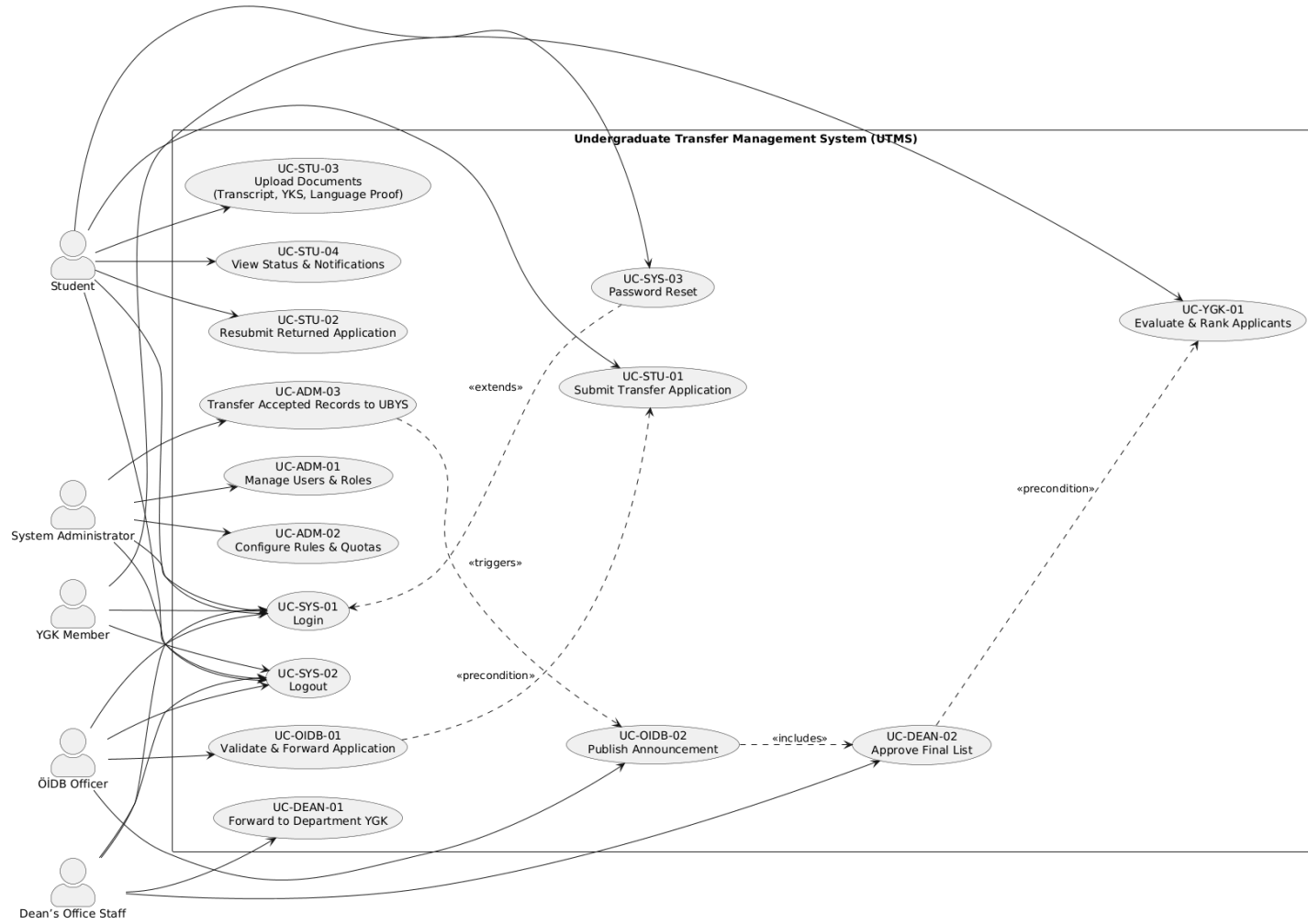
- **Student Portal** – Application submission & tracking
- **ÖİDB Panel** – Validation & routing
- **Dean's Panel** – Forwarding to departments
- **YGK Panel** – Evaluation & ranking
- **Announcement Module** – Public results
- **System Administrator** – System management

UTMS - System Context Diagram



Product Context Diagram

UTMS - Overall Use Case Diagram



Product Use Case Diagram

2.2 Product Functions

- Online application creation, editing, and submission
- Secure document upload (transcript, YKS certificate, proficiency proof)
- Automated GPA conversion and eligibility rules
- Role-based forwarding and decision logging
- Ranking and list generation (primary / waitlist)
- Returned-application workflow for corrections
- Faculty Board approval and final publication

2.3 User Characteristics

Actor	Description / Responsibility
Student	Submits applications, uploads documents, tracks status.
ÖİDB Officer	Validates submissions, returns invalid ones, forwards eligible applications.
Dean's Office Staff	Assigns applications to relevant YGK.
YGK Member	Evaluates applications based on academic rules and ranks candidates.
System Administrator	Manages users, configuration, and database integrity.

2.4 Constraints

- Comply with IZTECH data-handling regulations and YÖK rules.
- Hosted on servers using HTTPS (SSL/TLS 1.3).
- Integration limited to internal UBYS API.
- Application window open only during announced periods.

2.5 Assumptions and Dependencies

- Users have active institutional credentials.
- A stable internet connection is available.
- Evaluation rules (minimum GPA, quota, scoring) are preconfigured.
- Faculty and department data exist in the system before each cycle.

3. Specific Requirements

3.1 Use Cases

3.1.1 General System Use Cases

3.1.1.1 Use Case 1: User Login

Use Case ID: UC-SYS-01

Use Case Name: User Login

Created By: TEAM 3

Date: 25.10.2025

Actor: Student, ÖİDB Officer, Dean's Office Staff, YGK Member

Description: Allows any registered actor to log into the system by authenticating their identity.

Preconditions:

1. The actor has a valid username and password.
 2. The actor has a device with internet access.
- Postconditions:
3. The system authenticates the actor.
 4. The system redirects the actor to their role-appropriate dashboard.

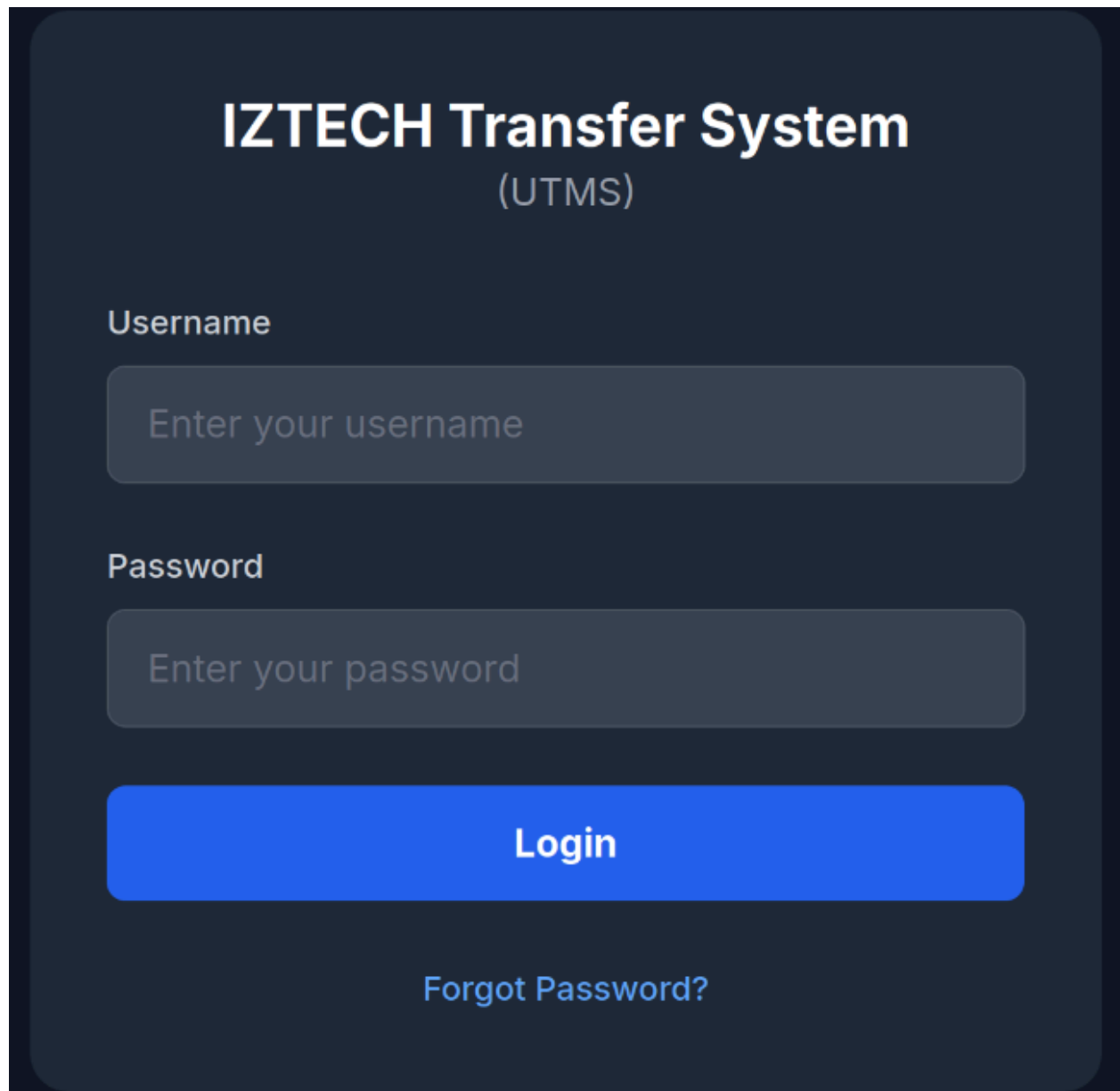
Normal Course:

1. The actor navigates to the system's main login page.
2. The system displays "Username" and "Password" input fields and a "Login" button.
3. The actor enters their credentials and clicks the "Login" button.
4. The system validates the entered information against database records. (See: EX.1, EX.2)
5. The system starts the actor's session and redirects them to their role-specific panel (e.g., Student Dashboard).

Exceptions:

- **UC-SYS-01.EX.1:** If the username or password is incorrect, the system displays the message "Incorrect username or password. Please try again."
- **UC-SYS-01.EX.2:** If the database connection fails, the system displays the message "System error. Please try again later."

3.1.1.1.1 User Interface (UC-SYS-01)



The image shows a login interface for the IZTECH Transfer System (UTMS). The interface is set against a dark blue background with rounded rectangular elements. At the top, the title "IZTECH Transfer System" is displayed in a large, bold, white font, with "(UTMS)" in a smaller font directly below it. Below the title, there are two input fields. The first is labeled "Username" in white text, and the second is labeled "Password" in white text. Both input fields have a light gray placeholder text that reads "Enter your username" and "Enter your password" respectively. Below the password field is a prominent blue button with the word "Login" in white text. At the bottom of the interface, there is a link that says "Forgot Password?" in a light blue color.

IZTECH Transfer System
(UTMS)

Username

Enter your username

Password

Enter your password

Login

[Forgot Password?](#)

3.1.1.2 Use Case 2: User Logout

Use Case ID: UC-SYS-02

Use Case Name: User Logout

Created By: TEAM 3

Date: 25.10.2025

Actor: Student, ÖİDB Officer, Dean's Office Staff, YGK Member

Description: Allows an authenticated actor to securely log out of the system.

Preconditions:

- The actor is logged into the system (active session).

Postconditions:

- The actor's session is terminated.
- The system redirects the actor to the login page.

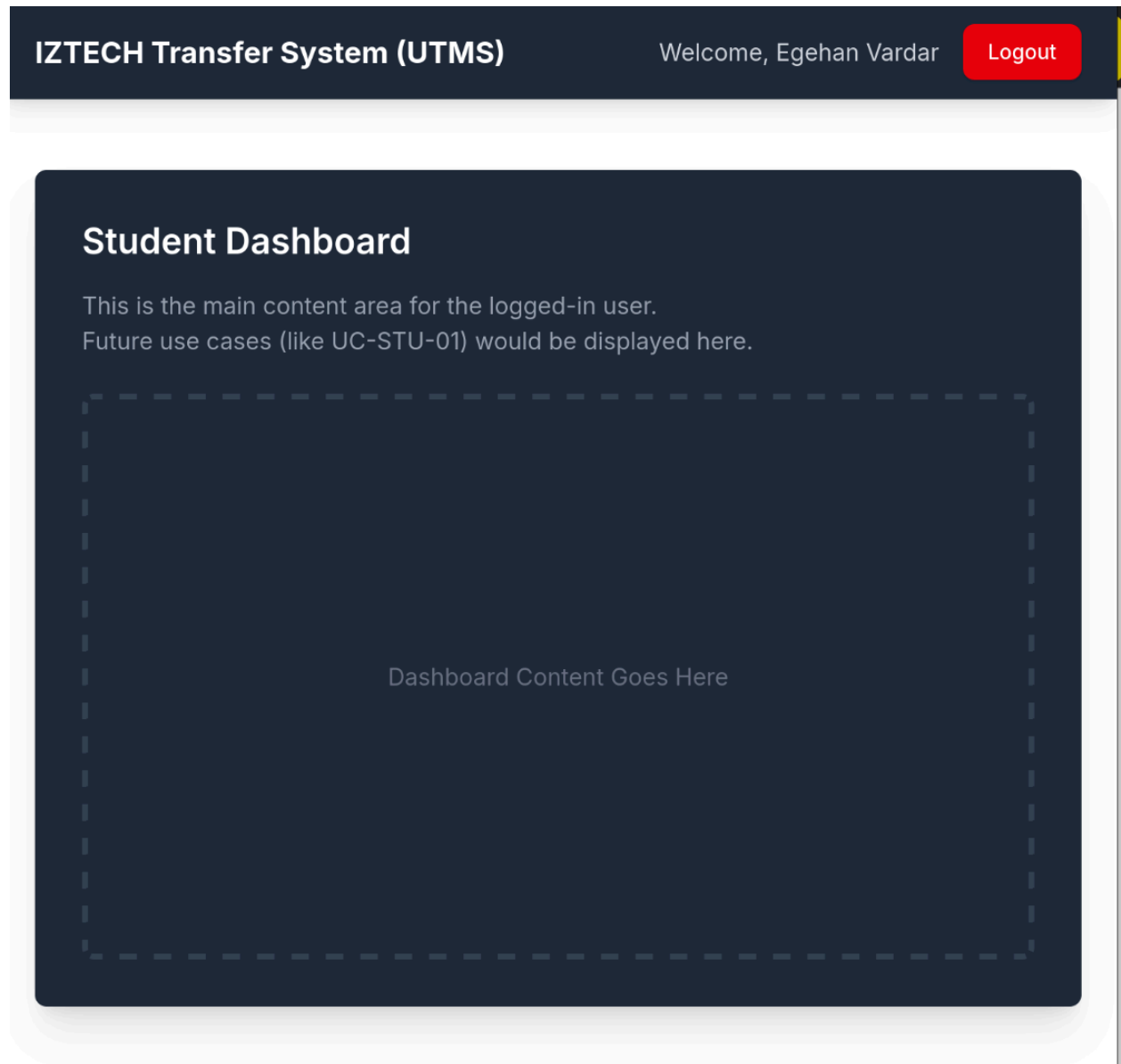
Normal Course:

1. The actor clicks the "Logout" button.
2. The system terminates and invalidates the session.
3. The system redirects the actor to the login page.

Exceptions:

- UC-SYS-02.EX.1: If a logout request fails due to a session timeout, the system automatically clears session cookies and prompts re-login.

3.1.1.1.2 User Interface (UC-SYS-02)



3.1.1.3 Use Case 3: Password Reset

Use Case ID: UC-SYS-03

Use Case Name: Password Reset

Created By: TEAM 3

Date: 31.10.2025

Actor: Student, ÖİDB Officer, Dean's Office Staff, YGK Member, System Administrator

Description: Registered UTMS users can securely reset forgotten passwords via "Forgot Password" using institutional email verification. The system validates the email, generates a one-time token, and allows the user to set a new password that meets security requirements.

Preconditions

1. The user has an existing registered account in UTMS.
2. The user has access to their institutional email address.

Postconditions

1. The user's old password is invalidated.
2. The system updates the user's credentials in the authentication database.
3. A confirmation email is sent to the user confirming the password change.

Normal Course (Main Flow)Other Use Case: Successful Password Reset

1. The user clicks "Forgot Password" on the login page.
2. The system displays the password reset form requesting an institutional email.
3. The user enters their institutional email address and clicks "Submit."
4. The system validates the email format and checks if it exists in the database.
5. If valid, the system generates a unique reset token and stores it with an expiration time (10 minutes).
6. The system sends a password reset email containing a secure link to the user's email.
7. The user clicks the reset link in the email.
8. The system verifies the token validity and opens the "Reset Password" page.
9. The user enters a new password and confirms it.
10. The system validates the password strength.
11. The user clicks "Save New Password."
12. The system updates the password (hashed and salted), invalidates the token, and confirms success.
13. A confirmation email is sent stating, "Your password has been successfully changed."

Alternative Flows

UC-SYS-03.AC.1 – Invalid Email Entered

- If the user provides an unregistered or improperly formatted email, the system will display the message: "Email not found. Please enter a valid institutional email."

UC-SYS-03.AC.2 – Token Expired or Invalid

- If the reset link is invalid or has expired, the system will display the message: "Reset link is invalid or has expired. Please request a new one."

UC-SYS-03.AC.3 – Weak Password

- If the new password fails to meet the specified policy criteria, the system will highlight the problem and require corrections before permitting submission.

Exception Flows

UC-SYS-03.EX.1 – Email Delivery Failure

- If an SMTP or network failure prevents the system from sending the reset email, it will log the error and display the message: "Unable to send reset email. Please try again later or contact the administrator."

UC-SYS-03.EX.2 – Database Error During Update

- If the password update is unsuccessful, the system will log the error, keep the existing credentials, and display the message: "Password reset could not be completed. Please try again later."

Assumptions

- The system's email service is active and configured.
- The user's institutional email is valid and accessible.
- Token expiration duration is configurable by the system administrator.

IZTECH Transfer System

(UTMS)

Forgot Password?

Enter your institutional email address below, and we'll send you a link to reset your password.

Institutional Email

e.g., egehanvardar@iyte.edu.tr

Send Reset Link

[← Back to Login](#)

3.1.2 Student Use Cases

3.1.2.1 Use Case 1: Submit Transfer Application

Use Case ID: UC-STU-01

Use Case Name: Submit Transfer Application

Created By: TEAM 3

Date: October 2025

Actor: Student

Description: The student submits a transfer application by entering personal and academic information, uploading required documents, and submitting it online.

Preconditions:

- The student is logged in (UC-SYS-01).
- The application window is open.

Postconditions:

- The application is recorded and assigned a tracking ID.
- The application is routed to the ÖİDB review queue.

Normal Course:

1. The student opens the “Transfer Application” page.
2. The system displays input fields for T.C. ID, GPA, and YKS Score.
3. The student fills all required fields.
4. The student uploads required documents (Transcript, YKS Result, Language Proof).
5. The system validates file formats and size. (See: AC.1)
6. The student clicks the “Submit Application” button.
7. The system saves the application, assigns a Tracking ID, and routes it to ÖİDB.
8. A confirmation message appears: “Your application has been successfully received. Tracking ID: UTMS-2025-XXXX.”

Alternative Course:

- UC-STU-01.AC.1: If validation fails (e.g., missing field or wrong file type), the system highlights the error and prompts correction before resubmission.

Exceptions:

- UC-STU-01.EX.1: If the database or server is unavailable, the system displays “System error. Please try again later.”

Assumptions:

- The student has stable internet access.

3.1.2.2.1 User Interface (UC-STU-01)

Transfer Application Form

IZTECH Transfer System (UTMS)

1. Personal & Academic Info

2. Document Upload

Full Name	T.C. Kimlik Numarası *
Egehan Vardar	11111111111111
Current University *	Current Program *
IYTE	SEDS
Overall GPA *	Grading System *
4	4.0 Scale
ÖSYM (YKS) Score *	Score Type *
500	SAY

Next (Document Upload) →

Transfer Application Form

IZTECH Transfer System (UTMS)

1. Personal & Academic Info

2. Document Upload

Transcript * (PDF Only, Max 5MB)

Browse...

No file selected.

ÖSYM Result Certificate * (PDF/JPEG, Max 2MB)

Browse...

No file selected.

Course Descriptions (Optional) (PDF, Max 10MB)

Browse...

No file selected.

☐ I confirm that the information and documents I have uploaded are correct. *

← Back

Submit Application

3.1.2.2 Use Case 2: Resubmit Returned Application

Use Case ID: UC-STU-02

Use Case Name: Resubmit Returned Application

Created By: TEAM 3

Date: October 2025

Actor: Student

Description: Enables a student to edit and resubmit an application previously returned by ÖİDB.

Preconditions:

- Application status is “Returned.”
- The correction reason is displayed to the student.

Postconditions:

- The application status changes to “Resubmitted.”

Normal Course:

1. The student logs in and accesses their returned application.
2. The system displays the return reason entered by ÖİDB.
3. The student corrects data or uploads revised documents.
4. The student clicks “Resubmit.”
5. The system validates and routes the application back to ÖİDB.

Exceptions:

- UC-STU-02.EX.1: If upload or validation fails, the system displays an error and prevents submission.

Assumptions:

- The correction note provided by ÖİDB clearly specifies the issue.

3.1.2.2.2 User Interface (UC-STU-02)

Resubmit Application

Your application (ID: UTMS-2025-1234) was returned.

Reason for Return (from ÖİDB):

"Your uploaded ÖSYM Result Certificate is blurry and unreadable. Please upload a clear, high-resolution PDF or JPEG of the document."

1. Personal & Academic Info

2. Document Upload

Full Name

Egehan Vardar (From System)

T.C. Kimlik Numarası *

12345678901

Current University *

IYTE

Current Program *

Physics

Overall GPA *

3.45

Grading System *

4.0 Scale

ÖSYM (YKS) Score *

480.75

Score Type *

SAY

Next (Document Upload) →

Resubmit Application

Your application (ID: UTMS-2025-1234) was returned.

Reason for Return (from ÖİDB):

"Your uploaded ÖSYM Result Certificate is blurry and unreadable. Please upload a clear, high-resolution PDF or JPEG of the document."

1. Personal & Academic Info

2. Document Upload

Transcript * (PDF Only, Max 5MB)

Previously uploaded: 'transcript_final.pdf' (You can upload a new file to replace it)

Browse...

No file selected.

ÖSYM Result Certificate * (PDF/JPEG, Max 2MB)

Previously uploaded: 'osym_blurry.jpg' (Re-upload required)

Browse...

No file selected.

Course Descriptions (Optional) (PDF, Max 10MB)

(No file previously uploaded)

Browse...

No file selected.

☐ I confirm that the *new* information and documents I have uploaded are correct. *

← Back

Resubmit Application

3.1.3 ÖİDB (Student Affairs Office) Use Cases

3.1.3.1 Use Case 1: Manage and Forward Applications

Use Case ID: UC-OIDB-01

Use Case Name: Manage and Forward Applications

Created By: TEAM 3

Date: October 2025

Actor: ÖİDB Officer

Description: The ÖİDB officer reviews submitted applications, verifies integrity, and forwards eligible ones to the appropriate faculty.

Preconditions:

- Applications are submitted by students.
- The ÖİDB officer is logged in.

Postconditions:

- Application status changes to “Forwarded to Faculty.”
- An audit record of the action is logged.

Normal Course:

1. The officer opens the “Incoming Applications” page.
2. The system lists new applications (ID, Name, Faculty).
3. The officer selects and reviews application details and documents.
4. If valid, clicks “Forward to Faculty.”
5. The system updates the status and creates a log entry.

Alternative Course:

- UC-OIDB-01.AC.1: If the application has missing or invalid documents, the officer selects “Return to Student for Correction,” adds a reason, and submits. The student is notified automatically.

Exceptions:

- UC-OIDB-01.EX.1: If a database timeout occurs, the system displays “Error: Application could not be forwarded. Please try again.”

3.1.3.1.1 User Interface (UC-OIDB-01)

ÖİDB - Incoming Applications Panel

Application Status

All

Faculty

All Faculties

Filter Applications

Application List

Egehan Vardar

TCKN: 12345678901

New

Ayşe Yılmaz

TCKN: 23456789012

New

Mehmet Öztürk

TCKN: 34567890123

Returned

Zeynep Kaya

TCKN: 99999999999

New

Zeynep Kaya

Koç University, Economics

TCKN

99999999999

Overall GPA

2.90 / 4.0

ÖSYM Score

450.50 (EA)

Target Faculty

Engineering

Documents

View Transcript (PDF)

View ÖSYM Certificate

Forward to Faculty

Return to Student

3.1.3.2 Use Case 2: Publish Announcement

Use Case ID: UC-OIDB-02

Use Case Name: Publish Announcement

Created By: TEAM 3

Date: October 2025

Actor: ÖİDB Officer

Description: Allows ÖİDB to upload final approved lists and publish results.

Preconditions:

- Dean's Office approval has been received.

Postconditions:

- Final results are published to the Announcement Module.
- Students are notified via email and portal.

Normal Course:

1. ÖİDB officer opens the "Announcements" section.
2. Uploads the final approved list.
3. Clicks "Publish Results."
4. The system updates status and sends notifications.

Exceptions:

- UC-OIDB-02.EX.1: If the system fails to send notifications, an alert appears: "Error: The notification could not be sent. Please try again."

Assumptions:

- Internet and database are available.

3.1.3.1.1 User Interface (UC-OIDB-02)

Publish New Announcement

Target Faculty *

Faculty of Engineering

Final Approved List * (PDF or XLSX)

[Browse...](#) No file selected.

☒ Send email and portal notifications to students.

[Publish Results](#)

Publication History

FACULTY	FILE NAME	PUBLISHED DATE
No results have been published yet.		

3.1.4 Dean's Office Use Cases

3.1.4.1 Use Case 1: Forward to Department Commission

Use Case ID: UC-DEAN-01

Use Case Name: Forward to Department Commission

Created By: TEAM 3

Date: October 2025

Actor: Dean's Office Staff

Description: Transfers applications received from ÖİDB to the respective departmental YGK for evaluation.

Preconditions:

- Applications have been received from ÖİDB.
- Relevant YGK exists for each department.

Postconditions:

- Applications appear in YGK evaluation queue.

Normal Course:

1. Dean's Office staff opens "Faculty Queue."
2. Selects applications.
3. Chooses target YGK from dropdown.
4. Clicks "Forward to YGK."
5. The system confirms and updates the status to "Under Review."

Exceptions:

- UC-DEAN-01.EX.1: If no YGK is assigned, the system displays "Error: No YGK defined for this department."

Assumptions:

- All departments and YGK members are active in the system.

3.1.4.1.1 User Interface (UC-DEAN-01)

Dean's Office - Application Routing Panel

Faculty of Engineering

Incoming Applications (from ÖİDB)

☐ Ahmet Yılmaz (Computer Eng.)

New

☐ Zeynep Kaya (Mechanical Eng.)

New

☒ Mehmet Öztürk (Computer Eng.)

New

Assign Selected:

Computer Engineering YGK

Forward to YGK

3.1.4.2 Use Case 2: Approve Final List

Use Case ID: UC-DEAN-02

Use Case Name: Approve Final List

Created By: TEAM 3

Date: October 2025

Actor: Dean's Office Staff

Description: Allows the Dean's Office to review, approve, or return the final ranked list prepared by the YGK before publication.

Preconditions:

- The YGK has finalized the evaluation and ranking list.
- The approved list has been sent to the Dean's Office.

Postconditions:

- Approved lists are transmitted to the ÖİDB for publication.
- Audit logs are recorded for the approval action.

Normal Course:

1. The Dean's Office user opens the "Final Lists" panel.
2. The system displays department-level ranked lists (Primary, Waitlist, Rejected).
3. The user reviews the lists for accuracy and compliance.
4. The user selects **Approve** or **Return for Revision**.
5. If approved, the system records the approval and sends the data to the ÖİDB queue.
6. A confirmation message appears: "List approved and sent to ÖİDB."

Alternative Course:

- UC-DEAN-02.AC.1: If minor corrections are needed, the Dean's Office can attach a comment and send the list back to the YGK for revision before approval.

Exceptions:

- UC-DEAN-02.EX.1: If the database connection fails during submission, the system displays: "Error: Approval could not be saved. Please try again."

Assumptions:

- The Dean's Office staff has appropriate permissions and digital signature access to approve results.

3.1.4.1.1 User Interface (UC-DEAN-02)

Dean's Office - Final List Approval

Finalized Lists

Computer Engineering Pending

Mechanical Engineering (Simulate Error) Pending

Computer Engineering - Final List Review

Asil Liste (Primary List)

Rank	Name	GPA	YKS Score
1	Mehmet Öztürk	3.85 / 4.0	512.00

Yedek Liste (Waitlist)

Rank	Name	GPA	YKS Score
1	Ahmet Yılmaz	3.45 / 4.0	480.75

Reddedilenler (Rejected)

Name	GPA	Reason
Zeynep Kaya	2.90 / 4.0	GPA below 3.0

Approve and Send to ÖİDB

Return to YGK for Revision

3.1.5 YGK (Transfer Commission) Use Cases

3.1.5.1 Use Case 1: Evaluate Applications

Use Case ID: UC-YGK-01

Use Case Name: Evaluate Applications

Created By: TEAM 3

Date: October 2025

Actor: YGK Member

Description: The YGK evaluates applications based on predefined eligibility and ranking rules, and generates final decisions.

Preconditions:

- Applications received from Dean's Office.
- Evaluation rules configured (see Section 4.3).

Postconditions:

- Ranked list (Primary, Waitlist, Rejected) generated.

Normal Course:

1. The YGK member opens the "Evaluation Panel."
2. The system lists all assigned applications.
3. The system auto-checks eligibility (min GPA, valid YKS, English proof).
4. Member reviews documents manually.
5. Member confirms or overrides automatic eligibility.
6. After all applications are reviewed, member clicks "Finalize Ranking."
7. The system ranks accepted students according to GPA and quota and sends results to Dean's Office.

Alternative Course:

- UC-YGK-01.AC.1: If the YKS score cannot be verified automatically, the system flags "Manual Verification Required."

Exceptions:

- UC-YGK-01.EX.1: If GPA conversion data is missing, the system displays "Error: GPA conversion rule not found. Please evaluate manually."

Assumptions:

- All evaluation rules are active in the system configuration.

3.1.5.1.1 User Interface (UC-YGK-01)

YGK Evaluation Panel

Computer Engineering

Please review all 4 pending application(s) before finalizing.

STUDENT NAME	GPA	YKS SCORE	AUTO-STATUS	FINAL DECISION	REVIEW
Mehmet Öztürk	3.85 / 4.0	512.00 (SAY)	Pre-Eligible	Accept	Review
Zeynep Kaya	2.90 / 4.0	475.50 (EA)	Not Eligible	Pending	Review
Ahmet Yılmaz	3.45 / 4.0	480.75 (SAY)	Pre-Eligible	Pending	Review
Ayşe Yılmaz	3.60 / 4.0	490.00 (SAY)	Manual Check	Pending	Review
Ali Veli	3.20 / 4.0	470.00 (SAY)	Error	Pending	Review

Generate Final Ranking and Send to Dean's Office

YGK Evaluation Panel

Computer Engineering

STUDENT NAME	GPA	YKS SCORE	AUTO-STATUS	FINAL DECISION	REVIEW
Mehmet Öztürk	3.85 / 4.0	512.00 (SAY)	Pre-Eligible	Accept	Review
Zeynep Kaya	2.90 / 4.0	475.50 (EA)	Not Eligible	Pending	Review
Ahmet Yılmaz	3.45 / 4.0	480.75 (SAY)	Pre-Eligible	Pending	Review
Ayşe Yılmaz					Review
Ali Veli					Review

Review: Ayşe Yılmaz

Student Information

TCKN:
56789012345Current University:
Boğaziçi UniversityCurrent Program:
PhysicsÖSYM (YKS) Score:
490.00 (SAY)Overall GPA:
3.60 / 4.0

Auto-Check Note

GPA 3.60 (Requirement Met), YKS Score 490.00 (Manual Verification Required). Please verify document.

Document Viewer

View Transcript (PDF)

View ÖSYM Result Certificate (PDF)

View Course Descriptions (PDF)

Final Decision

Pending

Cancel

Save Decision

3.1.6 Administrator Use Cases

3.1.6.1 Use Case 1: Manage Rules & Users

Use Case ID: UC-ADM-01

Use Case Name: Manage Rules & Users

Created By: TEAM 3

Date: October 2025

Actor: System Administrator

Description: Enables the system administrator to manage user accounts, roles, and predefined eligibility and evaluation rules in the UTMS Admin Console.

Preconditions:

- The administrator is authenticated and authorized with admin privileges.

Postconditions:

- Updated configuration and user data are saved and immediately applied.
- An audit log entry is created for each modification.

Normal Course:

1. The Administrator logs into the Admin Console.
2. The system displays management modules:
 - **User Management** (create, edit, deactivate accounts)
 - **Rule Management** (update predefined rules PR-01 to PR-24)
 - **Access Control (RBAC)** configuration
 - **System Monitoring** (logs, backups, performance stats)
3. The administrator selects a module (e.g., “Eligibility Rules”).
4. Edits or adds configuration values (e.g., GPA threshold = 2.50).
5. Clicks “Save Changes.”
6. The system validates input, updates database tables, and logs the action.
7. A confirmation message appears: “Changes saved successfully.”

Alternative Course:

- UC-ADM-01.AC.1: If invalid data (e.g., negative GPA threshold) is entered, the system displays: “Invalid input. Please enter a valid numerical value.”

Exceptions:

- UC-ADM-01.EX.1: If database or configuration file access fails, the system shows: “Error: Unable to save configuration. Please check the system logs.”

Assumptions:

- Only administrators with proper authorization tokens can modify rule definitions.
- Backup processes are in place before configuration changes are applied.

3.1.5.1.1 User Interface (UC-ADM-01)

IZTECH Transfer System (UTMS)

Welcome, System Administrator

Logout

Admin Console

Rule Management

User Management

Access Control (RBAC)

System Monitoring

Eligibility Rules

PR-01: Minimum GPA (4.0 Scale)

2.50

PR-03: Minimum TOEFL Score

70

Scoring & Ranking Rules

PR-08: Default Department Quota

10

Validation Rules

PR-11: Max File Size (MB)

5

PR-11: Allowed File Types

.pdf, .jpg, .jpeg

Set to 999 to simulate a save error (EX.1)

Save Changes

IZTECH Transfer System (UTMS)

Welcome, System AdministratorLogout

Admin Console

Rule Management

User Management

Access Control (RBAC)

System Monitoring

User Management

Add New User

Done

Name	Email	Role	Status	Actions
Zeynep Kaya	zeynep.kaya@iztech.edu.tr	ÖİDB Officer	Active	Edit
Ahmet Yilmaz	ahmet.yilmaz@iztech.edu.tr	YGK Member	Active	Edit

IZTECH Transfer System (UTMS)

Welcome, System AdministratorLogout

Admin Console

Rule Management

User Management

Access Control (RBAC)

System Monitoring

Access Control (RBAC)

Manage role permissions (Placeholder)

Admin Console

Rule Management User Management Access Control (RBAC) System Monitoring

System Monitoring (Logs)

```
[2025-10-31 10:01:15] INFO: User 'admin_user' logged in.
[2025-10-31 10:02:45] INFO: Student 'egehan.vardar' submitted application UTMS-2025-1234.
[2025-10-31 10:05:22] INFO: ÖİDB 'zeynep.kaya' forwarded application UTMS-2025-1234 to Faculty.
[2025-10-31 10:06:10] INFO: Dean 'dean_staff' forwarded application UTMS-2025-1234 to 'Computer Engineering YGK'.
[2025-10-31 10:10:50] INFO: YGK 'ahmet.yilmaz' updated decision for UTMS-2025-1234 to 'Accept'.
[2025-10-31 10:12:00] INFO: YGK 'ahmet.yilmaz' finalized list for 'Computer Engineering'.
[2025-10-31 10:15:30] INFO: Dean 'dean_staff' approved list for 'Computer Engineering'.
[2025-10-31 10:17:00] INFO: ÖİDB 'zeynep.kaya' published results for 'Computer Engineering'.
```

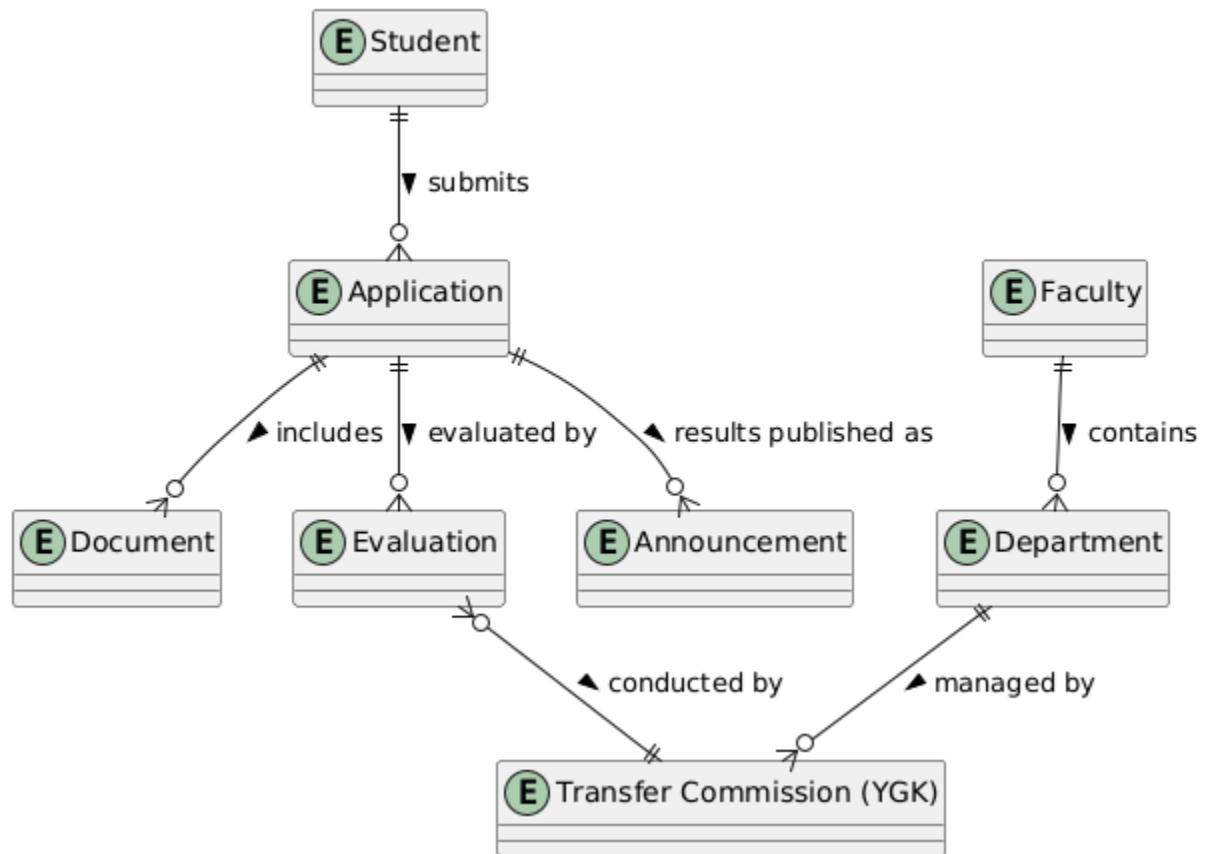
3.2 External Interfaces

The UTMS provides separate, role-based graphical interfaces that are accessible through a secure web browser.

Interface	Description	Main Components / Features
Student Portal	Allows students to submit, view, and track their transfer applications.	- Login/Logout page- Application Form (with validation and upload fields)- Status Dashboard- Notifications & Result View
ÖİDB Officer Panel	Used by Student Affairs officers to validate and forward applications, manage audit logs, and publish announcements.	- Incoming Applications Table- Return/Forward Buttons- Comment Entry Box- Result Publishing Module
Dean's Office Panel	Used to assign applications to departmental commissions and approve final results.	- Faculty Queue- YGK Assignment Dropdown- Approval Button- Audit Log View
YGK Evaluation Panel	Enables Transfer Commission members to perform eligibility checks, scoring, and ranking.	- Application Review Page- Eligibility Validation Section- Auto-Scoring & Ranking Functions- Finalization Button
Announcement Module (Public)	Displays published results for students and the public.	- Search by Student ID or Department- Downloadable Result Lists- Sorting & Filtering Options
Administrator Console	Restricted to system administrators for configuration and user management.	- User Management- Access Control Settings (RBAC)- Rule Editor (GPA, YKS thresholds, quotas)- Database & Log Monitoring

3.3 Logical Database Requirements (ER Diagram)

UTMS - Conceptual Entity Relationship Diagram



4. General Quality and Design Requirements

4.1 Performance Requirements

The system shall maintain responsiveness and reliability under expected workloads. Performance requirements are defined to ensure fast user interaction and efficient data processing.

Requirement ID	Description	Target / Measurement	Verification Method
PRF-01	Page Load Time	All user-facing pages (Login, Dashboard, Application Form) shall load in ≤ 3 seconds under 100 concurrent users .	Performance Test
PRF-02	Data Submission Time	Application submissions and document uploads shall complete in ≤ 5 seconds .	Test
PRF-03	Database Query Response	Common queries (e.g., status lookup, list retrieval) shall return results in ≤ 2 seconds .	Test
PRF-04	System Throughput	The system shall handle a minimum of 1500 simultaneous active sessions without degradation exceeding 10% response delay .	Load Test
PRF-05	Availability	System uptime shall be $\geq 99.5\%$ during business hours and $\geq 99\%$ overall .	Monitoring / Audit
PRF-06	Data Backup & Recovery	Database backup shall occur nightly , and recovery time objective (RTO) shall be ≤ 2 hours .	Inspection

4.2 Other Requirements

4.2.1 Security Requirements

UTMS must ensure the confidentiality, integrity, and availability of data in compliance with institutional and national data protection policies.

Requirement ID	Description	Target / Standard	Verification Method
SEC-01	Role-Based Access Control (RBAC)	Access permissions must be enforced by user role (Student, ÖİDB, Dean, YGK, Admin).	Test
SEC-02	Authentication	Login shall use secure session management and token-based authentication (OAuth 2.0 + JWT).	Inspection
SEC-03	Password Storage	All passwords shall be hashed and salted using PBKDF2 or bcrypt with minimum 10 iterations.	Code Review
SEC-04	Data Encryption	All data in transit via HTTPS (TLS 1.3) and data at rest encrypted with AES-256.	Security Audit
SEC-05	Audit Logging	All user actions (login, submission, approval, publication) must be timestamped and stored in immutable audit logs.	Inspection
SEC-06	Session Timeout	User sessions expire after 30 minutes of inactivity.	Test
SEC-07	Intrusion Detection	The system shall log unauthorized access attempts and notify	Test

		administrators.	
SEC-08	Backup Security	Backup files encrypted and accessible only by admin-level users.	Inspection
SEC-09	Password Complexity	Passwords shall be at least 8 characters long, include uppercase, lowercase, numbers, and special characters.	Code Review

4.2.2 Usability Requirements

The UTMS interface must ensure clarity, accessibility, and consistency across platforms to minimize user errors and training needs.

Requirement ID	Description	Target / Metric	Verification Method
USE-01	Responsiveness	The system shall maintain full functionality on desktop, tablet, and mobile devices.	Test
USE-02	Accessibility Compliance	UI design must conform to WCAG 2.1 Level AA standards.	Inspection
USE-03	Ease of Navigation	$\geq 90\%$ of test participants must successfully complete application submission in ≤ 10 minutes without assistance.	Usability Test
USE-04	Error Feedback	All validation errors shall display user-friendly messages (e.g., "File exceeds 5 MB limit").	Test
USE-05	Interface Consistency	All pages follow a unified design system (color scheme, typography, button layout).	Inspection
USE-06	Browser Compatibility	Fully functional on Chrome version 142, Firefox ESR 128	Test

4.2.3 Maintainability Requirements

UTMS must be modular, configurable, and easily maintainable to support evolving policies and workflows.

Requirement ID	Description	Target / Standard	Verification Method
MAI-01	Modularity	Code shall be organized into separate modules (Authentication, Application, Evaluation, Announcement).	Code Review
MAI-02	Configuration Management	All system rules (GPA threshold, quotas, deadlines) stored in editable configuration files or database tables — not hard-coded.	Inspection
MAI-03	Documentation	All modules documented with inline comments and updated API specifications.	Review
MAI-04	Error Handling	Centralized error-logging system captures and classifies all runtime errors for diagnosis.	Test
MAI-05	Update & Deployment	System updates deployable with zero downtime via CI/CD pipeline.	Test
MAI-06	Maintainability Index	Code maintainability index ≥ 80 (based on static analysis tools).	Analysis

4.2.4 Scalability and Portability

Requirement ID	Description	Target / Standard	Verification Method
SCP-01	Scalability	The system shall support increased user load (up to 2000 concurrent users) with $\leq 15\%$ performance loss.	Load Test
SCP-02	Portability	UTMS shall operate on both Linux and Windows servers with no code modification.	Test
SCP-03	Cloud Readiness	System deployable in containerized environment (Docker/Kubernetes) for future scaling.	Analysis
SCP-04	Integration Flexibility	REST API endpoints compatible with existing university systems (UBYS, YÖKSİS, ÖSYM).	Test

4.3 Pre-Defined Rules

4.3.1 Eligibility Rules

Rule ID	Description	Example Condition
PR-01	Minimum GPA rule	Student's GPA (converted to 4.0 scale) ≥ 2.50
PR-02	Enrollment type rule	Transfer type $\in \{\text{Internal, External, Central Placement}\}$
PR-03	English proficiency rule	TOEFL ≥ 70 OR IZTECH English Proficiency Exam = Pass
PR-04	Disciplinary record check	Must have "No active disciplinary penalty"
PR-05	Application period validity	Current date within announced transfer window
PR-06	Course completion rule	≥ 1 completed semester with GPA available

4.3.2 Scoring & Ranking Rules

Rule ID	Description	Formula / Criteria
PR-07	Composite Transfer Score Calculation	Transfer Score = (GPA * 0.5) + (YKS Score * 0.5)
PR-08	Quota Rule (Per Department)	Accept $\leq$ department_quota
PR-09	Tie-breaking Rule	Higher YKS Score $\rightarrow$ Higher GPA $\rightarrow$ Earliest Submission Time
PR-10	Ranking Cutoff Rule	Generate “Primary List” = Top N; “Waitlist” = Next $N \times 0.5$

4.3.3 Validation Rules

Rule ID	Description	Validation Check
PR-11	File Format Validation	Uploaded Transcript must be .PDF $\leq$ 5 MB
PR-12	Mandatory Field Validation	T.C. No, Faculty, and GPA fields cannot be empty
PR-13	Data Integrity Check	Numeric fields only in GPA/YKS fields
PR-14	Document Authenticity	Compare uploaded YKS code with ÖSYM web API
PR-15	Duplicate Prevention	Reject multiple submissions from same student ID

4.3.4 Evaluation & Decision Rules

Rule ID	Description	Action
PR-16	Auto-Eligibility Determination	Mark “Eligible” or “Not Eligible” in YGK Panel
PR-17	Quota Enforcement	System blocks acceptance beyond defined quota
PR-18	Auto-Ranking Finalization	System generates “Asil/Yedek” (Primary/Waitlist) list
PR-19	Decision Lock Rule	YGK cannot edit ranking after final approval
PR-20	Audit Logging Rule	Every YGK action (decision/save) is timestamped

4.3.5 Announcement & Publication Rules

Rule ID	Description	Example
PR-21	Result Visibility Rule	Students can only view their own status until official publication
PR-22	Publication Format Rule	Results exported as .PDF and .XLSX , stored in Announcement Module
PR-23	Notification Trigger Rule	Send email/SMS upon status change in 5 minutes
PR-24	Archival Rule	Archive published results after 90 days

4.4 Requirements Traceability Matrix

FR ID	Functional Requirement	Linked NFR(s)	Linked Use Case(s)	Verification Method
FR-01	The system shall allow users (Students, ÖİDB, Dean, YGK) to log in using secure institutional credentials.	SEC-02, SEC-06, USE-01	UC-SYS-01	Test / Inspection
FR-02	The system shall allow users to securely log out and invalidate session tokens.	SEC-02, SEC-06	UC-SYS-02	Test
FR-03	The system shall enable students to create and submit a transfer application with required data and documents.	USE-03, PRF-01, PRF-02	UC-STU-01	Test
FR-04	The system shall validate uploaded files (type, size, authenticity).	PR-11–PR-14, SEC-05	UC-STU-01	Test / Inspection
FR-05	The system shall allow students to view and correct returned applications.	USE-03, MAI-02	UC-STU-02	Test
FR-06	ÖİDB officers shall review submitted applications and either return or forward them to faculties.	SEC-01, MAI-01	UC-OİDB-01	Inspection / Test
FR-07	Dean's Office shall forward received applications to the appropriate YGK department.	SEC-01, MAI-02	UC-DEAN-01	Test
FR-08	YGK members shall evaluate applications and perform automated eligibility checks (GPA, YKS, proficiency).	PR-01–PR-06, PRF-03	UC-YGK-01	Test / Analysis
FR-09	The system shall generate ranked candidate lists (Primary/Waitlist) based on scoring rules.	PR-07–PR-10, PRF-04	UC-YGK-01	Test
FR-10	Dean's Office shall approve or return final lists before publication.	SEC-05, MAI-02	UC-DEAN-02	Inspection / Test
FR-11	ÖİDB shall publish approved results and send notifications to students.	SEC-08, USE-04	UC-OİDB-02	Test
FR-12	The system shall record all user actions (submission, forwarding, evaluation, approval) in immutable audit logs.	SEC-05, MAI-04	All UCs	Inspection / Audit

FR-13	The system shall provide public access to final results via Announcement Module with search/filter functions.	USE-06, SCP-04	UC-OIDB-02	Test
FR-14	The Administrator shall manage users, roles, and system configuration (GPA thresholds, quotas).	SEC-01, MAI-01, MAI-02	UC-ADM-01	Inspection / Test
FR-16	The system shall integrate with UBYS and ÖSYM APIs for data validation.	SCP-04, SEC-02	UC-YGK-01	Integration Test
FR-17	The system shall prevent duplicate submissions by the same student ID.	PR-15, SEC-05	UC-STU-01	Test
FR-18	The system shall restrict access based on user role (RBAC).	SEC-01	All UCs	Security Test
FR-19	The system shall provide error feedback messages for invalid inputs or failed actions.	USE-04	All UCs	Usability Test
FR-20	The system shall generate audit and performance reports for administrative review.	PRF-05, MAI-04	UC-ADM-01	Analysis